

# Extreme Subgroup Effects

*Under a Uniform Treatment Benefit*

Section 6.6.4 — High-Risk Subgroup Analysis

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# The Question

*Statistical and clinical motivation*

## STATISTICAL OBJECTIVE

When a treatment has a perfectly uniform benefit across all patients, how often do standard subgroup analyses produce extreme-looking results by chance alone?

## WHY IT MATTERS

Regulatory reviewers see forest plots with apparently alarming subgroup hazard ratios or upper confidence bounds. Are these real signals — or sampling noise from small denominators?

*Answer: extreme results are routine in small subgroups — clinical rationale provides no statistical protection.*

# The Simulation Setup

*Real-data DGM with a perfectly uniform treatment effect*

1

## Source data

GBSG breast cancer trial: N = 686 patients, 299 events

2

## Weibull AFT model

Fit to the source data; build a super-population (N = 5,000)

3

## Uniform treatment effect

True hazard ratio = 0.70 for every patient — no heterogeneity, by construction

4

## Censoring calibration

uniroot-based search matches simulated censoring to the source data

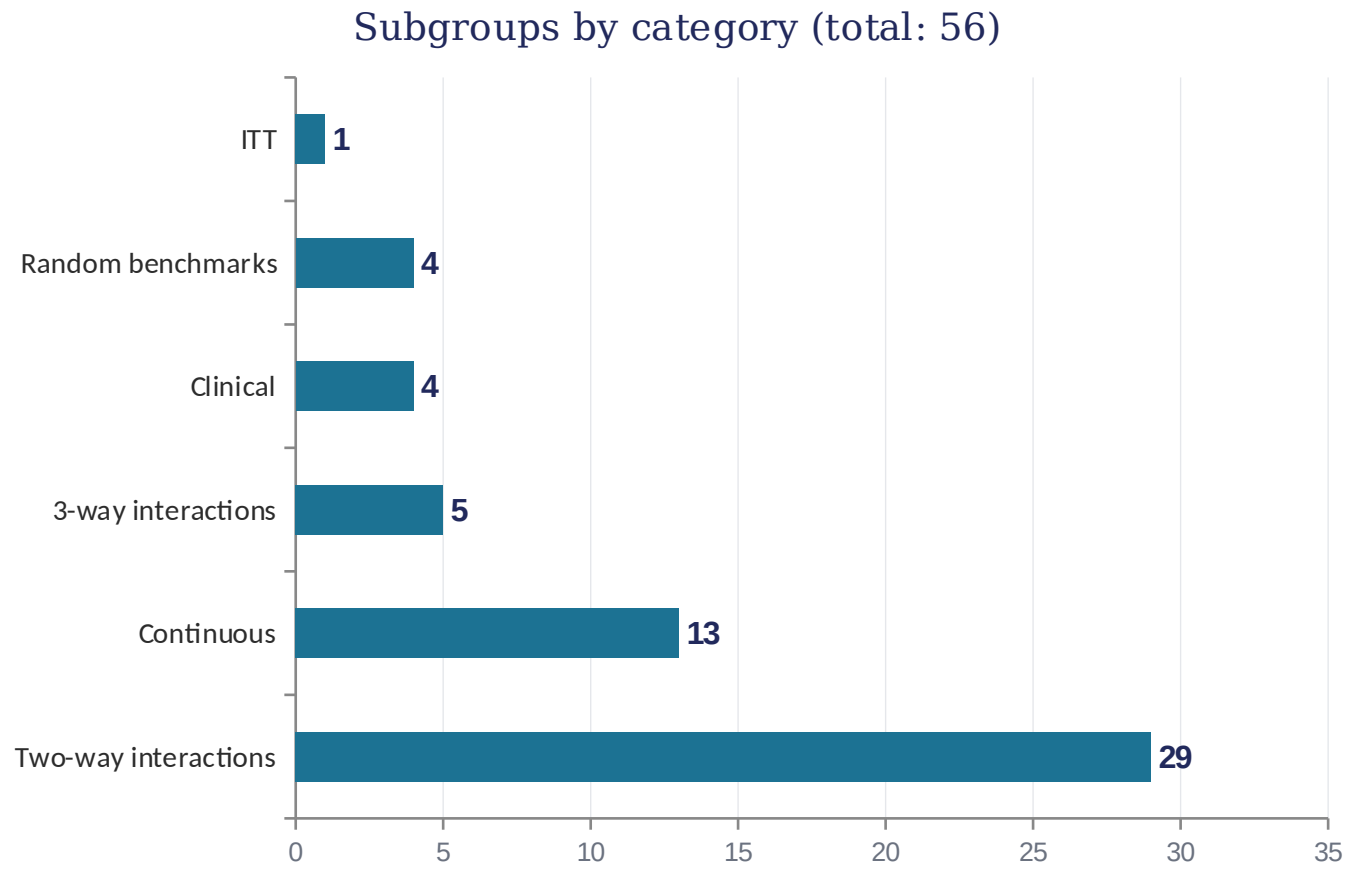
5

## 10,000 synthetic trials

Each trial: 686 patients × 56 pre-defined subgroups × Cox stratified by grade

# Subgroups Under Study

56 subgroups examined in every simulated trial



## Why so many — and what each adds

**ITT (1)** Full trial — anchors the comparison

**Clinical (4)** Standard categorical splits (meno, grade)

**Continuous (13)** Median / quartile cut-points on age, size, nodes, PGR, ER

**Two-way (29)** Pairwise clinical x biomarker combinations

**3-way (5)** Three-way intersections — progressively smaller N

**Random (4)** Pure-noise benchmarks: random60 / 40 / 20 / 15

# Section 6.6.4 — High-Risk Subgroups

*Filtering to subgroups most prone to extreme upper bounds*

## **FILTER RULE**

**$\Pr( \text{UB(HR)} \geq 2.0 ) \geq 10\%$  across 10,000 simulated trials**

*All Patients (ITT) included for comparison regardless of whether it meets the filter.*

### **Two views, one set of subgroups**

**UB(HR) panel** asks: how alarming can the upper confidence bound look by chance?

**HR panel** asks: how misleading can the point estimate itself be?

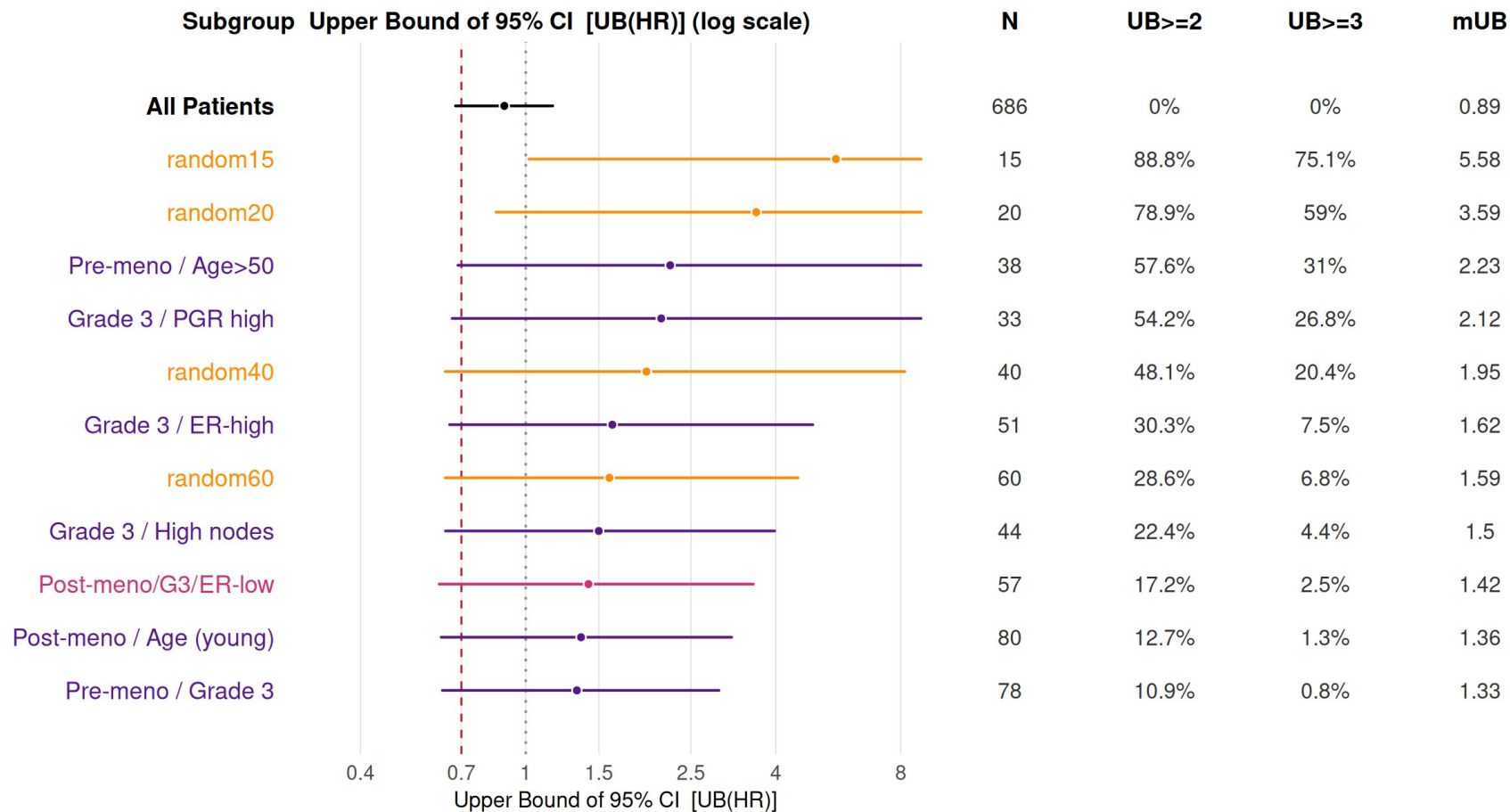
### **11 subgroups + ITT comparator**

Random benchmarks (random15 / 20 / 40 / 60) lead the list — pure sampling noise.

Clinical and biomarker-defined subgroups of similar size are statistically indistinguishable from them.

# UB(HR) Distribution — Section 6.6.4 Panel A

What a safety reviewer sees: the upper 95% confidence bound, across simulations

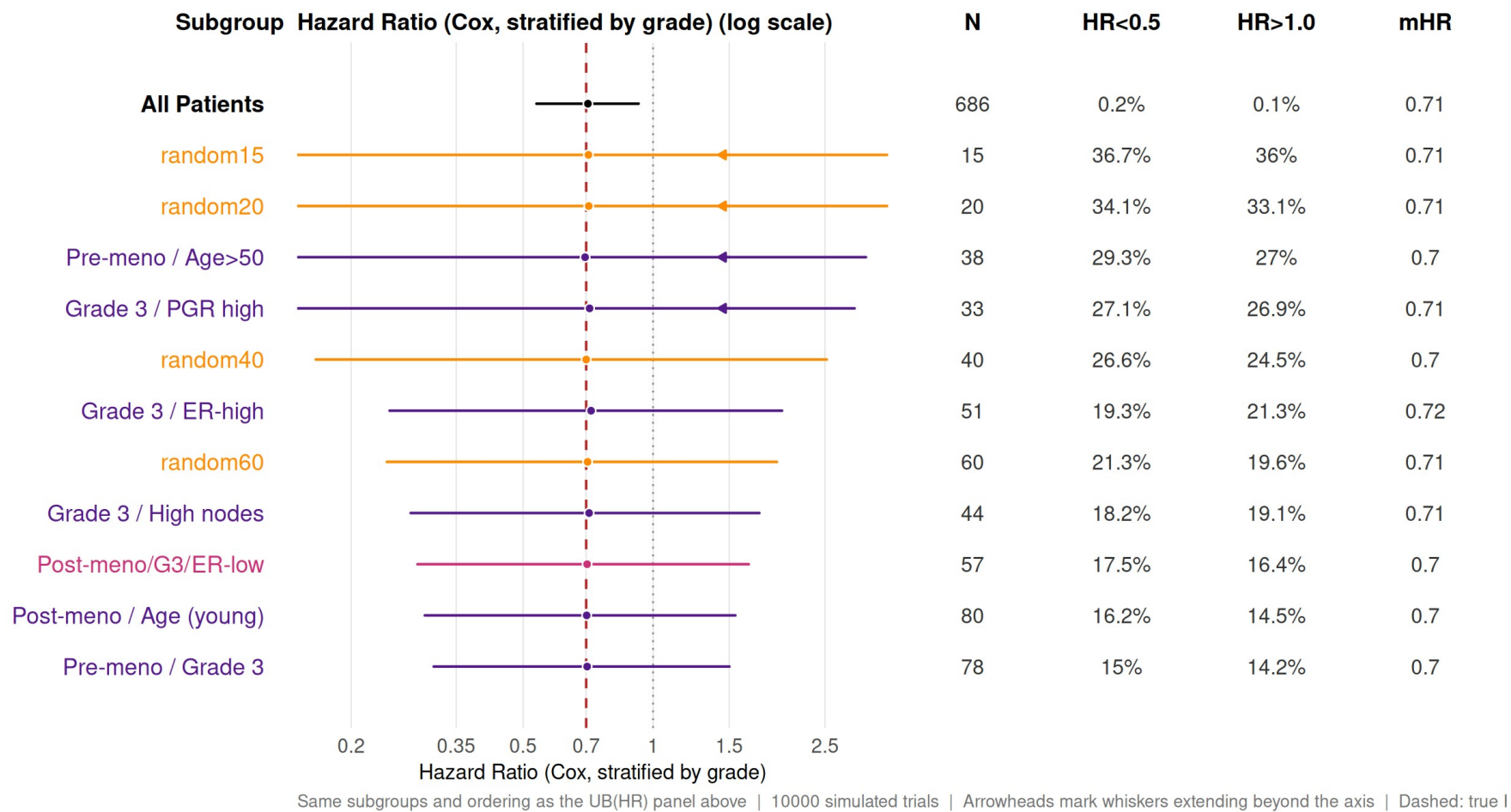


Filter: Pr(UB(HR) >= 2.0) >= 10% | ITT included for comparison | 10000 simulated trials | Arrowheads mark whiskers extending beyond the axis | Dotted: l

- random15 / random20 produce UB >= 2 in 75-89% of trials and UB >= 3 in 59-75%.
- Clinical / biomarker subgroups of similar size (Pre-meno/Age>50, Grade 3/PGR high, ...) show indistinguishable distributions.
- Arrowheads mark whiskers whose 99th percentile extends beyond the visible axis.

# HR Distribution — Section 6.6.4 Panel B

Same subgroups, point-estimate view: how misleading can the headline HR be?



- Every row's median HR is approximately 0.70 — the Cox model is unbiased even at N = 15.
- But 27-37% of trials in small subgroups produce HR < 0.5 (looks like a huge benefit); 25-36% produce HR > 1.0 (looks like harm).
- All this variability is sampling noise — the true HR is 0.70 by construction.

# Key Takeaways

*Reading the UB and HR panels together*

## 1 Extremes are routine in small subgroups

Under a uniform HR = 0.70, random15 trials produce UB(HR) > 5 about three-quarters of the time. Same true effect for everyone — the apparent heterogeneity is sampling noise.

## 2 Clinical rationale != protection

Pre-meno/Age>50 (N approx 38) behaves like random40 (N approx 40). A defined subgroup of size N has the same chance distribution as a random pick of N.

## 3 The Cox model is working correctly

Median HR is approximately 0.70 across every row. The wide CIs accurately reflect uncertainty — the problem is interpretation, not the statistics.

## 4 Simulation-based calibration is the fix

Without a null benchmark, no reviewer can quantify how 'extreme' UB = 3.5 in a 40-patient subgroup actually is. The random\* anchors provide that calibration.



# Resources

## Package

[github.com/larry-leon/forestsearch](https://github.com/larry-leon/forestsearch)

## Documentation

[larry-leon.github.io/forestsearch/](https://larry-leon.github.io/forestsearch/)

## Vignette

[larry-leon.github.io/forestsearch/articles/extreme\\_subgroups.html](https://larry-leon.github.io/forestsearch/articles/extreme_subgroups.html)

## Install

```
pak::pak("larry-leon/forestsearch")
```